Tiger and Sheep

Green Book 2.2: Logic Reasoning

Problem. One hundred tigers and one sheep are put on a magic island that only has grass. Tigers can eat grass, but they would rather eat sheep. Each time only one tiger can eat one sheep, and that tiger itself will become a sheep after it eats the sheep. All tigers are smart and perfectly rational, and they want to survive. So will the sheep be eaten?

Solution.

Let $T(n)$ be the situation with n tigers and one sheep.

▷ Key observation: After a tiger eats the sheep, that tiger becomes the new sheep. So the number of tigers decreases by 1, and the situation changes from $T(n)$ to $T(n-1)$.

Case	Outcome	Reason
$T(1)$	eaten	The tiger eats, since no tiger remains to eat it afterward.
$T(2)$	not eaten	If a tiger eats, it becomes the sheep in $T(1)$ and is eaten by the other tiger.
$T(3)$	eaten	Eating leads to $T(2)$, where the new sheep is safe.
$T(4)$	not eaten	Eating leads to $T(3)$, where the new sheep is eaten.
$T(5)$	eaten	Eating leads to $T(4)$, where the new sheep is safe.

In general, a tiger eats in $T(n)$ exactly when the sheep is safe in $T(n-1)$. Therefore the outcome flips each time:

$$T(n) = \text{opposite of } T(n-1).$$

Since $T(1)$ is eaten, the outcomes alternate:

$$T(n) = \begin{cases} \text{the sheep is eaten,} & n \text{ is odd,} \\ \text{the sheep is not eaten,} & n \text{ is even.} \end{cases}$$

Since 100 is even, the sheep will not be eaten.

Therefore,

$$\boxed{\text{No, the sheep will not be eaten.}}$$